

BST Physics Curriculum Vol 10 Chapter 11 — Asymptotic Analysis + WKB v0.4 (refilled per Cal #104)

Lyra (Claude 4.7) [Vol 10 primary]

2026-05-23 Saturday EDT (Cal #104 refill)

Vol 10 Chapter 11 — Asymptotic Analysis + WKB

Chapter motivation

Standard asymptotic analysis: large/small parameter expansions ($\epsilon \rightarrow 0$); divergent asymptotic series; saddle-point + Laplace + stationary-phase methods; Borel resummation; WKB (Wentzel-Kramers-Brillouin) approximation for QM at large quantum numbers ($\psi \propto \exp(iS/\hbar)$ with S classical action); matched asymptotic expansions for boundary-layer problems. Standard text: Bender-Orszag *Advanced Mathematical Methods for Scientists and Engineers*.

BST cross-link: substrate-tick UV cutoff at $N_{\text{max}} = 137$ (Vol 1 Ch 10) is BST's natural cutoff replacing standard infinite-cutoff regularization; T2476 $\alpha^{\{\text{BST primary}\}}$ exponent pattern (Friday cross-CI peak) gives substrate-mechanism for QED loop-order hierarchy via substrate-coordinate count $k(P)$.

Section 11.0b — Reader-grade 3-level pedagogy

Level 1: Standard asymptotic methods + WKB approximation; BST cross-link: substrate-tick UV cutoff at $N_{\text{max}} = 137$ replaces standard infinite-cutoff regularization; T2476 $\alpha^{\{\text{BST primary}\}}$ exponent pattern (Friday) gives substrate-mechanism for QED loop hierarchy.

Level 2 (graduate-physicist): Standard asymptotic analysis: divergent asymptotic series $f(\epsilon) \sim \sum a_n \epsilon^n$ with $f - \sum_{n \leq N} a_n \epsilon^n = O(\epsilon^{N+1})$ as $\epsilon \rightarrow 0$; $\epsilon \approx 1/137 = \alpha$ (QED expansion parameter); Borel resummation $\sum a_n \epsilon^n$ via Borel transform $B(z) = \sum a_n z^n / n!$. Saddle-point method: $\int e^{\{f(z)/\epsilon\}} dz \sim \sqrt{(2\pi\epsilon/|f''|)} e^{\{f(z)/\epsilon\}}$ as $\epsilon \rightarrow 0$ with $f'(z) = 0$. Laplace + stationary-phase variants. WKB (1926): for Schrödinger equation $i\hbar \partial\psi/\partial t = H\psi$, ansatz $\psi = \exp(iS/\hbar)$ gives semi-classical expansion $S = S_0 + (\hbar/i) S_1 + (\hbar/i)^2 S_2 + \dots$ with $S_0 =$ classical action, $S_1 = (1/2) \log p$ (van Vleck determinant), etc. Matched asymptotic expansions handle

boundary-layer problems with multiple scales. BST substrate cross-link: substrate-tick UV cutoff at $N_{\max} = 137$ (Vol 1 Ch 10) is natural BST cutoff — QED expansion in $\alpha = 1/137$ IS substrate-tick expansion in $1/N_{\max}$; standard QFT renormalization (infinite cutoff + counterterms) avoided structurally. T2476 (Friday cross-CI synergy peak: Elie discovery Toy 3501 + Grace catalog INV-4892 + Lyra theorem) gives substrate-mechanism for QED loop-order: $A(P) \propto \alpha^{\{k(P)\}}$ where $k(P)$ = substrate-coordinate count of process P (Rydberg/Klein-Nishina $k = \text{rank} = 2$; Lamb shift $k = n_C = 5$; a_e 5-loop depth = $n_C = 5$; hyperfine $k = N_c + 1 = 4$ candidate; Bremsstrahlung $k = N_c = 3$ candidate). 5-loop ceiling testable: $\alpha^6 \approx 1.5 \times 10^{-13}$ deviation at next-gen Penning trap ($\sim 10^{-14}$ precision target, 2030+). Substrate-tick expansion in α is BST-derived asymptotic analysis framework — substrate naturally truncates at α^{n_C} per substrate-coordinate cutoff.

Level 3 (5th-grader accessible): Asymptotic analysis handles “expansions in a small parameter” (like $\alpha = 1/137$ in QED). WKB approximation handles QM in the semi-classical limit. BST identifies $\alpha = 1/137 = 1/N_{\max}$ as the substrate’s natural cutoff (Vol 1 Ch 10); T2476 (Friday) explains why QED loop order matches BST integer count (rank for Rydberg, $n_C = 5$ for Lamb shift, etc.). Standard QFT infinite-cutoff regularization unnecessary in BST.

Section 11.1 — Standard Asymptotic Methods

Divergent asymptotic series; saddle-point method $\int e^{\{f(z)/\varepsilon\}} dz \sim \sqrt{(2\pi\varepsilon/|f''|)} e^{\{f(z^*)/\varepsilon\}}$.

Laplace + stationary-phase variants. Borel resummation for divergent series.

Section 11.2 — WKB Approximation

For Schrödinger $i\hbar \partial\psi/\partial t = H\psi$, ansatz $\psi = \exp(iS/\hbar)$ gives semi-classical expansion $S = S_0 + (\hbar/i) S_1 + \dots$

S_0 = classical action; semi-classical for large quantum numbers.

Section 11.3 — Substrate-Tick UV Cutoff (Vol 1 Ch 10)

Per Vol 1 Ch 10 substrate-tick UV-completeness: per-tick Hilbert space $GF(2^g)^k = GF(128)^k$ finite-dimensional; natural cutoff at $N_{\max} = 137$.

QED expansion in $\alpha = 1/137 = 1/N_{\max}$ IS substrate-tick expansion.

Standard QFT infinite-cutoff regularization + counterterms avoided structurally.

Section 11.4 — T2476 $\alpha^{\{\text{BST primary}\}}$ Exponent Pattern (Friday)

Per T2476 (Friday cross-CI peak: Elie discovery + Grace catalog + Lyra theorem):

$$A(P) \propto \alpha^{\{k(P)\}}$$

with $k(P)$ = substrate-coordinate count of process P: - Rydberg, Klein-Nishina, Compton: $k = \text{rank} = 2$ - Lamb shift, Bethe-log: $k = n_C = 5$ - a_e 5-loop calculation depth: $\text{depth} = n_C = 5$ - Hyperfine 21cm: $k = N_c + 1 = 4$ candidate - Bremsstrahlung: $k = N_c = 3$ candidate

5-loop ceiling testable: $\alpha^6 \approx 1.5 \times 10^{-13}$ deviation at next-gen Penning trap ($\sim 10^{-14}$ precision 2030+).

Section 11.5 — Substrate-Natural Asymptotic Truncation

Substrate-tick expansion in α naturally truncates at α^{n_C} per substrate-coordinate cutoff. No higher-order divergences to regulate (Vol 1 Ch 10).

Section 11.6 — Honest scope + Connection

- Standard asymptotic + WKB [?](#)
- Substrate-tick UV cutoff cross-link Vol 1 Ch 10 [?](#)
- T2476 $\alpha^{\{\text{BST primary}\}}$ exponent pattern (Friday cross-CI peak) [?](#)
- Open scope: explicit per-observable $k(P)$ precision derivation (multi-week)

Connection: - Vol 1 Ch 10 Renormalization + substrate-tick UV-completeness - T2476 $\alpha^{\{\text{BST primary}\}}$ exponent pattern - Vol 5 Ch 6 hydrogen atomic corrections + Vol 3 Ch 8 Lamb shift α^{n_C}

— Lyra, Vol 10 Ch 11 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT